

Retail Analysis

2026-08-11

Introduction

This project analyzes historical retail sales from February of 2010 to November of 2012 and the different factors associated with changes in weekly retail sales. This specific data set focuses on holidays, markdowns, CPI (consumer price index), and other economic conditions related to consumer spending. This leads to the main research question:

What factors are associated with changes in weekly retail sales, and how do holidays and economic conditions influence retail sales?

The analysis combines three data sets: store, features, and sales. The Sales data set contains weekly sales by store and department. The Features data set contains information about economic conditions, weather, fuel prices, and promotional markdowns. The Stores data set contains information about store type and size.

Data Overview

```
head(sales)
```

##	Store	Dept	Date	Weekly_Sales	IsHoliday
## 1	1	1	05/02/2010	24924.50	FALSE
## 2	1	1	12/02/2010	46039.49	TRUE
## 3	1	1	19/02/2010	41595.55	FALSE
## 4	1	1	26/02/2010	19403.54	FALSE
## 5	1	1	05/03/2010	21827.90	FALSE
## 6	1	1	12/03/2010	21043.39	FALSE

```
head(features)
```

##	Store	Date	Temperature	Fuel_Price	Markdown1	Markdown2	Markdown3
## 1	1	05/02/2010	42.31	2.572	NA	NA	NA
## 2	1	12/02/2010	38.51	2.548	NA	NA	NA
## 3	1	19/02/2010	39.93	2.514	NA	NA	NA
## 4	1	26/02/2010	46.63	2.561	NA	NA	NA
## 5	1	05/03/2010	46.50	2.625	NA	NA	NA
## 6	1	12/03/2010	57.79	2.667	NA	NA	NA

##	Markdown4	Markdown5	CPI	Unemployment	IsHoliday
## 1	NA	NA	211.0964	8.106	FALSE
## 2	NA	NA	211.2422	8.106	TRUE
## 3	NA	NA	211.2891	8.106	FALSE
## 4	NA	NA	211.3196	8.106	FALSE
## 5	NA	NA	211.3501	8.106	FALSE
## 6	NA	NA	211.3806	8.106	FALSE

```
head(stores)
```

##	Store	Type	Size
## 1	1	A	151315
## 2	2	A	202307

```
## 3      3      B 37392
## 4      4      A 205863
## 5      5      B 34875
## 6      6      A 202505
```

```
str(sales)
```

```
## 'data.frame':    421570 obs. of  5 variables:
## $ Store      : int  1 1 1 1 1 1 1 1 1 1 ...
## $ Dept       : int  1 1 1 1 1 1 1 1 1 1 ...
## $ Date       : chr   "05/02/2010" "12/02/2010" "19/02/2010" "26/02/2010" ...
## $ Weekly_Sales: num  24924 46039 41596 19404 21828 ...
## $ IsHoliday  : logi  FALSE TRUE FALSE FALSE FALSE FALSE ...
```

```
str(features)
```

```
## 'data.frame':    8190 obs. of  12 variables:
## $ Store      : int  1 1 1 1 1 1 1 1 1 1 ...
## $ Date       : chr   "05/02/2010" "12/02/2010" "19/02/2010" "26/02/2010" ...
## $ Temperature: num  42.3 38.5 39.9 46.6 46.5 ...
## $ Fuel_Price : num  2.57 2.55 2.51 2.56 2.62 ...
## $ Markdown1  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ Markdown2  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ Markdown3  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ Markdown4  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ Markdown5  : num  NA NA NA NA NA NA NA NA NA NA ...
## $ CPI        : num  211 211 211 211 211 ...
## $ Unemployment: num  8.11 8.11 8.11 8.11 8.11 ...
## $ IsHoliday  : logi  FALSE TRUE FALSE FALSE FALSE FALSE ...
```

```
str(stores)
```

```
## 'data.frame':    45 obs. of  3 variables:
## $ Store: int  1 2 3 4 5 6 7 8 9 10 ...
## $ Type : chr   "A" "A" "B" "A" ...
## $ Size : int  151315 202307 37392 205863 34875 202505 70713 155078 125833 126512 ...
```

```
summary(sales)
```

```
##      Store      Dept      Date      Weekly_Sales
## Min.   : 1.0    Min.   : 1.00   Length:421570   Min.   : -4989
## 1st Qu.:11.0    1st Qu.:18.00   Class :character 1st Qu.: 2080
## Median :22.0    Median :37.00   Mode  :character Median : 7612
## Mean   :22.2    Mean   :44.26                      Mean   : 15981
## 3rd Qu.:33.0    3rd Qu.:74.00                      3rd Qu.: 20206
## Max.   :45.0    Max.   :99.00                      Max.   :693099
## IsHoliday
## Mode :logical
## FALSE:391909
## TRUE :29661
##
##
##
```

```
summary(features)
```

```
##      Store      Date      Temperature      Fuel_Price
## Min.   : 1    Length:8190   Min.   : -7.29   Min.   :2.472
```

```
## 1st Qu.:12   Class :character   1st Qu.: 45.90   1st Qu.:3.041
## Median :23   Mode  :character   Median : 60.71   Median :3.513
## Mean   :23                                     Mean  : 59.36   Mean   :3.406
## 3rd Qu.:34                                     3rd Qu.: 73.88   3rd Qu.:3.743
## Max.   :45                                     Max.   :101.95   Max.   :4.468
##
##      Markdown1      Markdown2      Markdown3      Markdown4
## Min.   : -2781     Min.   : -265.76   Min.   : -179.26   Min.   :  0.22
## 1st Qu.: 1578     1st Qu.:  68.88   1st Qu.:  6.60   1st Qu.: 304.69
## Median : 4744     Median :  364.57   Median :  36.26   Median : 1176.42
## Mean   : 7032     Mean   : 3384.18   Mean   : 1760.10   Mean   : 3292.94
## 3rd Qu.: 8923     3rd Qu.: 2153.35   3rd Qu.: 163.15   3rd Qu.: 3310.01
## Max.   :103185    Max.   :104519.54   Max.   :149483.31   Max.   :67474.85
## NA's   :4158     NA's   :5269     NA's   :4577     NA's   :4726
##      Markdown5      CPI      Unemployment      IsHoliday
## Min.   : -185.2     Min.   :126.1     Min.   : 3.684     Mode :logical
## 1st Qu.: 1440.8     1st Qu.:132.4     1st Qu.: 6.634     FALSE:7605
## Median : 2727.1     Median :182.8     Median : 7.806     TRUE :585
## Mean   : 4132.2     Mean   :172.5     Mean   : 7.827
## 3rd Qu.: 4832.6     3rd Qu.:213.9     3rd Qu.: 8.567
## Max.   :771448.1     Max.   :229.0     Max.   :14.313
## NA's   :4140     NA's   :585     NA's   :585
```

```
summary(stores)
```

```
##      Store      Type      Size
## Min.   : 1     Length:45     Min.   : 34875
## 1st Qu.:12     Class :character   1st Qu.: 70713
## Median :23     Mode  :character   Median :126512
## Mean   :23                                     Mean  :130288
## 3rd Qu.:34                                     3rd Qu.:202307
## Max.   :45                                     Max.   :219622
```

The Sales data set contains five variables. The weekly sales range from the negatives to over \$600,000 with the average being close to \$16,000. The data set also contains both holiday and non-holiday observations. Negative sales values are present in the data set and should be considered when interpreting the results because they may represent returns or other adjustments rather than ordinary retail sales.

The Features data set includes the most variables: Store (identification number), Date, Temperature (during the week), Fuel Price, Markdowns (up to five promotional markdowns during the week), CPI (Consumer Price Index), Unemployment Rate, and whether it was a holiday week. This data set is particularly important for the economic portion of this project because it includes CPI and unemployment which can be used to investigate whether other outside economic conditions are associated with the weekly retail spending.

The Stores data set contains 45 stores and identifies the category of the store, while Size measures the physical size of the store.

```
retail <- sales %>% left_join( features, by = c("Store", "Date", "IsHoliday") ) %>% left_join( stores, by = c("Store") )
```

After combining the data sets, the new retail data set contains sales information along with economic conditions, promotional information, and store characteristics.

```
retail$Date <- as.Date(retail$Date, "%d-%m-%Y")
colSums(is.na(retail))
```

```
##      Store      Dept      Date Weekly_Sales      IsHoliday      Temperature
##      0         0      421570         0         0         0
## Fuel_Price  Markdown1  Markdown2  Markdown3  Markdown4  Markdown5
```

```
##           0           270889           310322           284479           286603           270138
##           CPI Unemployment           Type           Size
##           0           0           0           0
```

```
retail <- retail %>% mutate( Markdown1 = replace_na(Markdown1, 0), Markdown2 = replace_na(Markdown2, 0)

retail <- retail %>% mutate(TotalMarkdown = Markdown1 + Markdown2 + Markdown3 + Markdown4 + Markdown5)
summary(retail$TotalMarkdown)
```

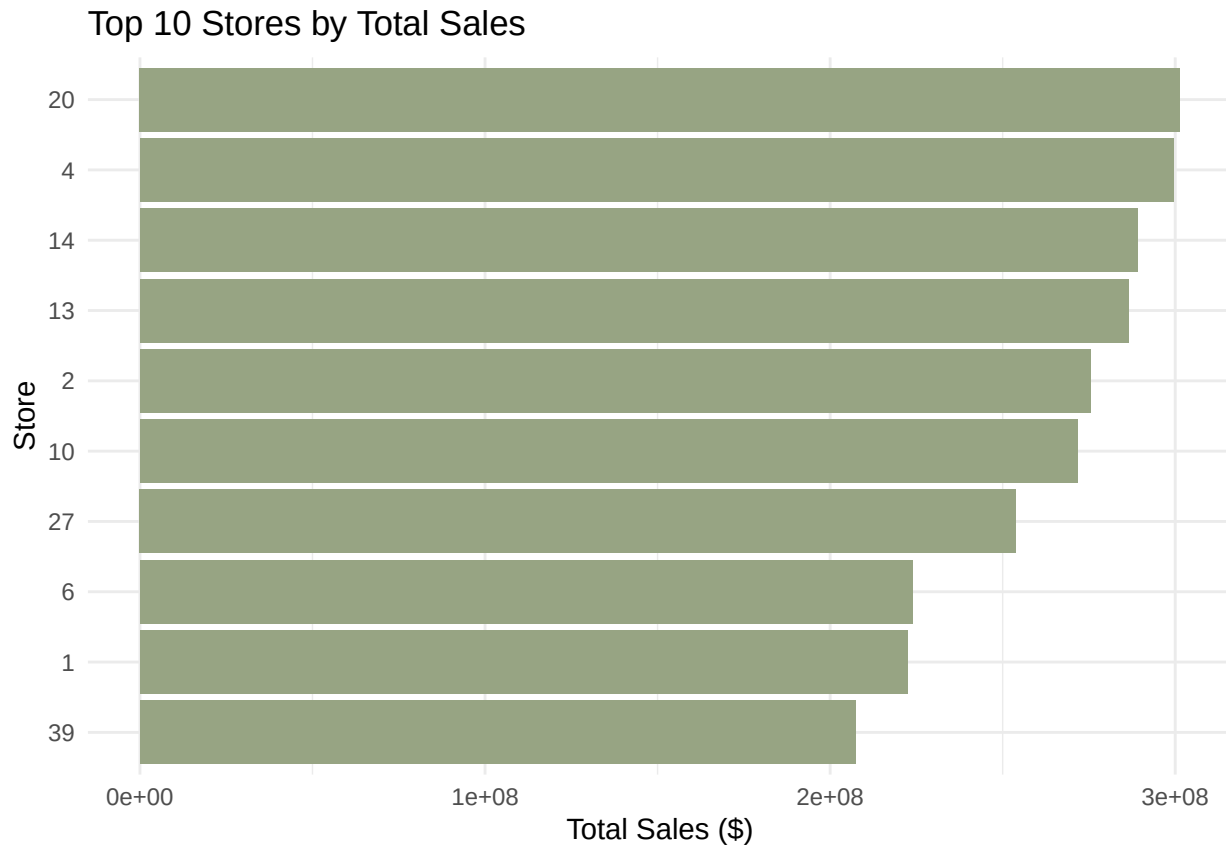
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##           0         0         0   6684   8075  160511
```

Checking through the data to convert the Date variable so that it can be used for time-based analysis and, if displayed incorrectly, it is fixed. Also, going through the variables in the Features data set to see if there are any missing variables, especially for the markdown variables. We put a 0 instead of NA which indicates that there was no markdown recorded that week.

The retail variable will be used later to investigate whether larger promotional markdowns are associated with higher weekly sales.

Plot Data Analysis

```
store_sales <- retail %>%
  group_by(Store) %>%
  summarize(TotalSales = sum(Weekly_Sales))
ggplot(store_sales,
  aes(x = reorder(Store, TotalSales),
    y = TotalSales)) +
  geom_col(fill = "#dce9f3") +
  coord_flip() +
  labs(
    title = "Total Sales by Store",
    x = "Store",
    y = "Total Sales"
  ) +
  theme_minimal()
```

However, with there being 45 stores, the number is too grand for the graph to interpret. Therefore, the analysis focuses on the ten stores with the highest total sales.

This graph shows which 10 stores generated the highest total sales during the period covered by the data set. The graph shows that Store 20 generated the most sales with Store 4 coming behind as a close second.

The differences between stores may reflect differences in store size, location, customer demand, or store type.

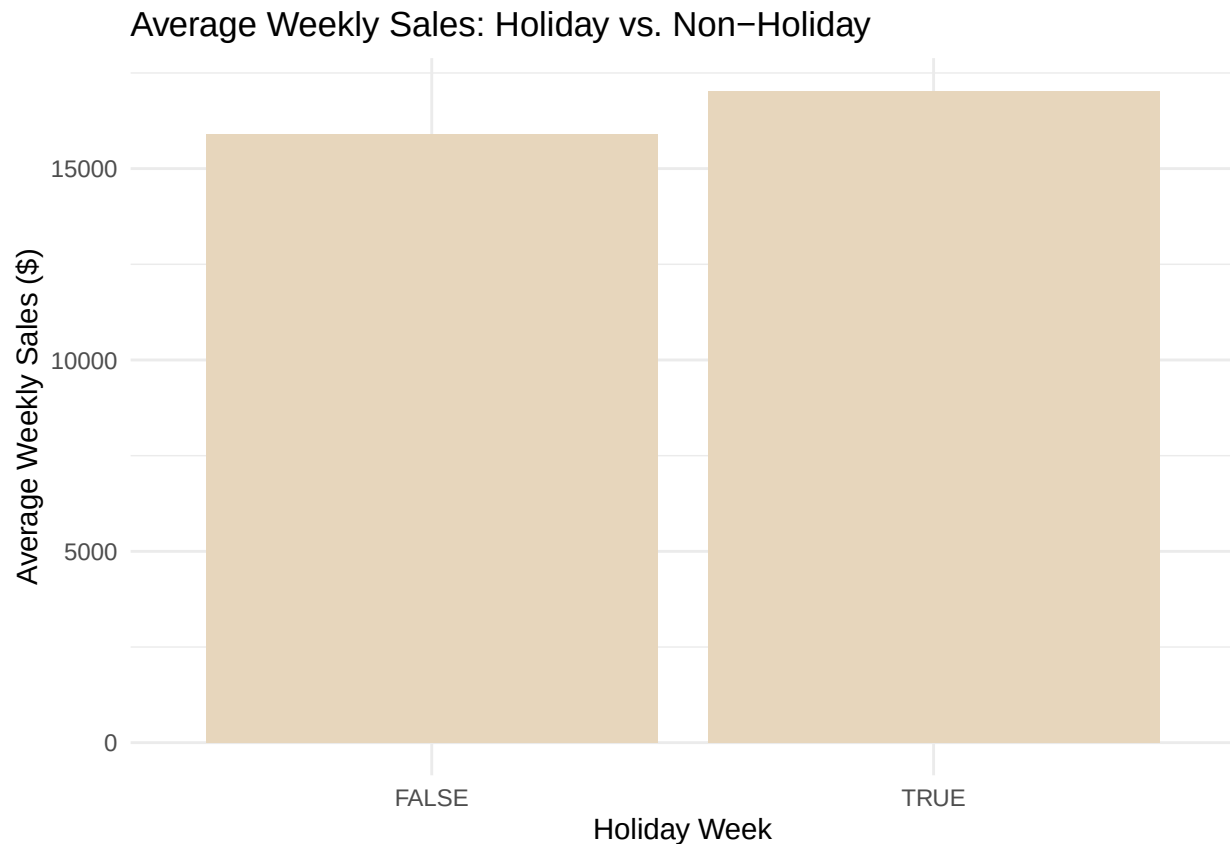
Are weekly sales higher during holiday weeks than during non-holiday weeks?

```
monthly_sales <- retail %>%
  mutate(Month = format(Date, "%Y-%m")) %>%
  group_by(Month) %>%
  summarize(Sales = sum(Weekly_Sales, na.rm = TRUE))

holiday_sales <- retail %>%
  group_by(IsHoliday) %>%
  summarize(
    AverageSales = mean(Weekly_Sales, na.rm = TRUE)
  )

ggplot(holiday_sales,
  aes(x = IsHoliday,
    y = AverageSales)) +
  geom_col(fill = "#e7d6bc") +
  labs(
    title = "Average Weekly Sales: Holiday vs. Non-Holiday",
    x = "Holiday Week",
    y = "Average Weekly Sales ($)"
  )
```

```
) +  
theme_minimal()
```



The IsHoliday variable identifies whether a particular week contains a major holiday.

The graph compares weekly sales during holiday and non-holiday weeks. It shows that holiday weeks have a higher average of weekly sales which suggests that major holidays are associated with increased retail activity.

However, this comparison alone does not prove that holidays are the sole reason for causing higher sales. Other factors, such as promotions or seasonal patterns, may also contribute to the difference.

Sales Overtime

```
sales <- sales %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
  
features <- features %>%  
  mutate(Date = as.Date(Date, format = "%d/%m/%Y"))  
  
retail <- sales %>%  
  left_join(  
    features,  
    by = c("Store", "Date", "IsHoliday")  
  ) %>%  
  left_join(  
    stores,  
    by = "Store"  
  )
```

```

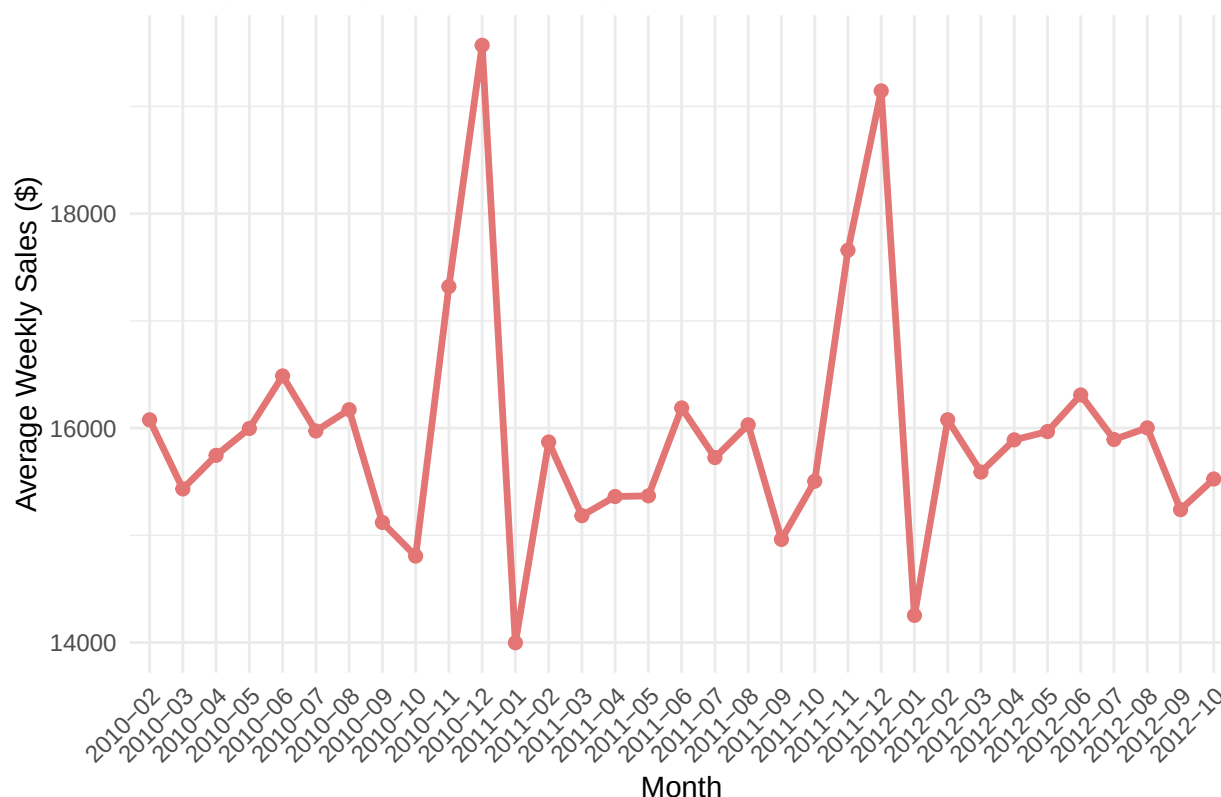
retail <- retail %>%
  mutate(
    Month = format(Date, "%Y-%m")
  )

monthly_sales <- retail %>%
  mutate(
    Month = format(Date, "%Y-%m")
  ) %>%
  group_by(Month) %>%
  summarize(
    AverageWeeklySales = mean(
      Weekly_Sales,
      na.rm = TRUE
    )
  )

ggplot(
  monthly_sales,
  aes(
    x = Month,
    y = AverageWeeklySales,
    group = 1
  )
) +
  geom_line(
    color = "#e37574",
    linewidth = 1.2
  ) +
  geom_point(
    color = "#e37574",
    size = 2
  ) +
  labs(
    title = "Average Weekly Retail Sales by Month",
    x = "Month",
    y = "Average Weekly Sales ($)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )

```


Average Weekly Retail Sales by Month



This analysis helps identify seasonal patterns in retail sales and whether certain periods experience higher or lower sales. The graph shows that the average weekly sales are relatively steady except for sudden peaks around November and December months and drops during January.

This suggests that the previous graph on how holiday weeks result in higher average weekly sales supports this graph because of the peaks in the line graph. However, the steep drops in the line graph can suggest the results in the holiday shopping because of factors like less customer traffic or drops in consumer spending.

Economic Factors

The Features data set allows a deeper dive in the sales analysis and investigate economic factors. The main factors include: CPI (consumer price index), Unemployment (unemployment rate), and Fuel Price (average price of fuel associated with the observation).

The main goal is to determine whether these variables are associated with differences in weekly retail sales.

Unemployment

Since the data set contains many observations, a random sample can be used for visualization so that the graph is easier to interpret.

```
set.seed(123)

sample_data <- retail %>%
  sample_n(5000)

ggplot(
  sample_data,
```

```

aes(
  x = Unemployment,
  y = Weekly_Sales
) +
geom_point(
  color = "#18fbff",
  alpha = 0.4
) +
geom_smooth(
  method = "lm",
  color = "#20b2aa"
) +
labs(
  title = "Unemployment and Weekly Retail Sales",
  x = "Unemployment Rate",
  y = "Weekly Sales ($)"
) +
theme_minimal()

```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
cor( retail$Unemployment, retail$Weekly_Sales, use = "complete.obs" )
```

```
## [1] -0.02586372
```

This analysis measures association rather than causation. The correlation measures the strength and direction of the relationship between unemployment and weekly sales.

Since the correlation is negative, it indicates that higher unemployment is associated with lower weekly sales.

CPI

This analysis examines whether changes in Consumer Price Index are associated with differences in weekly retail sales.

A relationship between CPI and sales could reflect changes in consumer prices and spending patterns.

```
ggplot(
  sample_data,
  aes(
    x = CPI,
    y = Weekly_Sales
  )
) +
  geom_point(
    color = "#c6c6e8",
    alpha = 0.4
  ) +
  geom_smooth(
    method = "lm",
    color = "#9896bb"
  ) +
  labs(
    title = "CPI and Weekly Retail Sales",
    x = "Consumer Price Index",
    y = "Weekly Sales ($)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
cor(
  retail$CPI,
  retail$Weekly_Sales,
  use = "complete.obs"
)
```

```
## [1] -0.02092134
```

The scatter plot shows that the correlation is extremely close to zero (horizontal line) which indicates that there is no linear relationship between a store's/region CPI and its weekly sales figures. This means that changes in CPI does not predict lower or higher weekly sales in this data set.

Fuel Prices

Fuel prices may influence consumer spending because transportation costs affect household budgets and travel behavior.

```
ggplot(
  sample_data, aes( x = Fuel_Price, y = Weekly_Sales
  )
) + geom_point( color = "#ff85a2", alpha = 0.4
) +
  geom_smooth( method = "lm", color = "#2ec4b6" ) +
  labs( title = "Fuel Prices and Weekly Retail Sales",
        x = "Fuel Price",
        y = "Weekly Sales ($)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The scatter plot shows that the completely flat regression line (teal) confirms that there is no linear correlation between fuel prices and weekly retail sales volume. All of the sales remain heavily concentrated below \$50,000 no matter the fuel price.

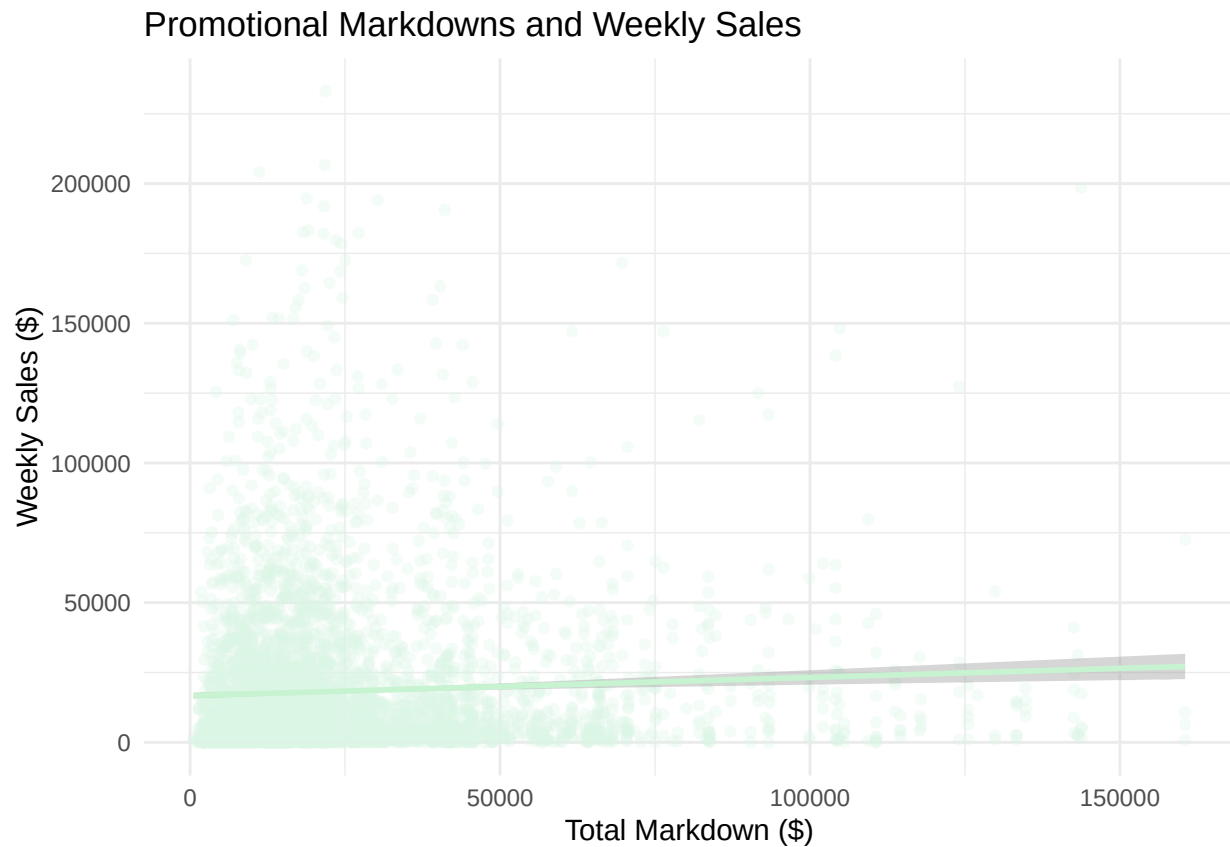
Markdowns

The data set contains five different markdown variables. A combined markdown variable was created to examine the overall amount of markdown activity.

```
retail <- retail %>%  
  mutate(  
    TotalMarkdown =  
      Markdown1 +  
      Markdown2 +  
      Markdown3 +  
      Markdown4 +  
      Markdown5  
  )  
  
markdown_sample <- retail %>% filter(TotalMarkdown > 0) %>%  
  sample_n(5000)  
  
ggplot(  
  markdown_sample,  
  aes(  
    x = TotalMarkdown,  
    y = Weekly_Sales  
  )  
) +
```

```
geom_point(
  color = "#dcf6e6",
  alpha = 0.3
) +
geom_smooth(
  method = "lm",
  color = "#c7f3d0"
) +
labs(
  title = "Promotional Markdowns and Weekly Sales",
  x = "Total Markdown ($)",
  y = "Weekly Sales ($)"
) +
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
cor(
  retail$TotalMarkdown,
  retail$Weekly_Sales,
  use = "complete.obs"
)
```

```
## [1] 0.08270748
```

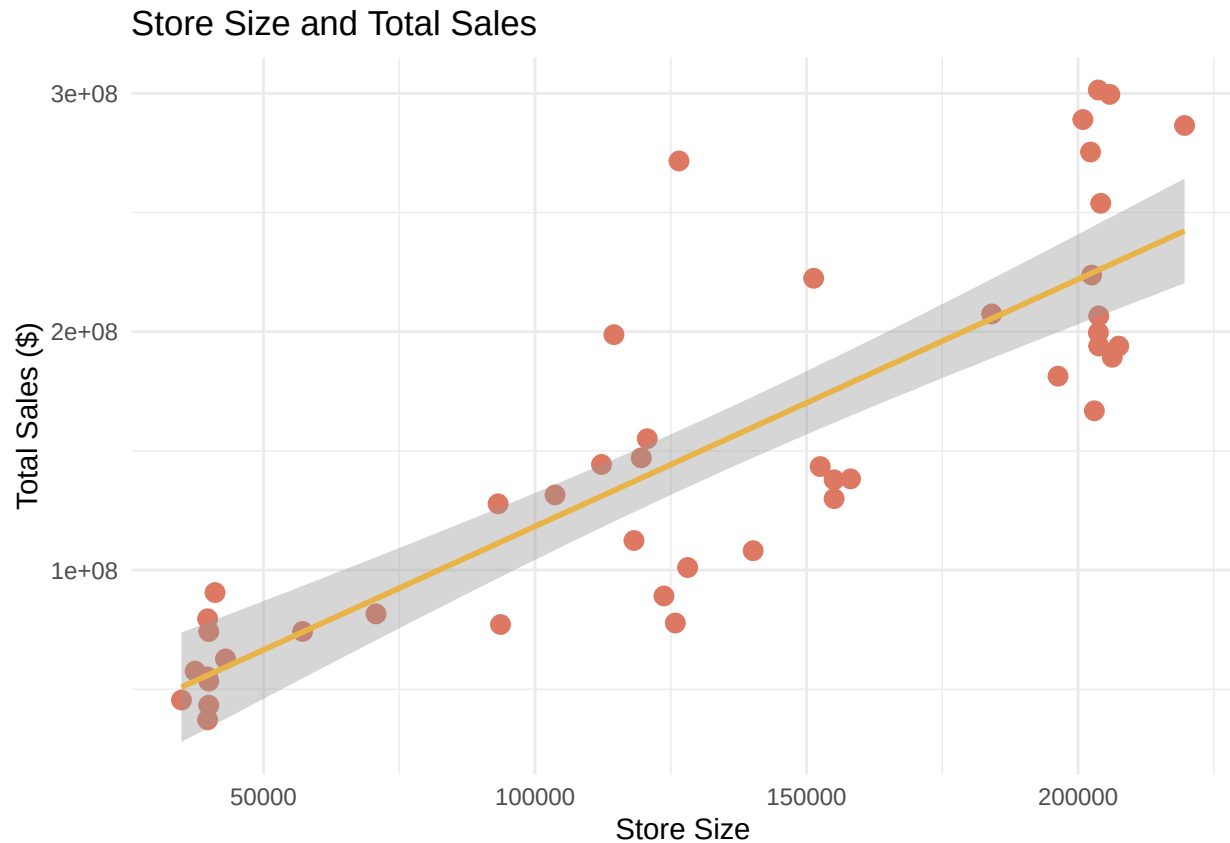
The upward-sloping trend line (with a shaded confidence interval band) shows that higher markdowns correspond to a negligible average increase in weekly sales.

Store Size and Sales

```
store_analysis <- retail %>%
  group_by(Store) %>%
  summarize(
    TotalSales = sum(Weekly_Sales, na.rm = TRUE)
  ) %>%
  left_join(
    stores,
    by = "Store"
  )

ggplot(
  store_analysis,
  aes(
    x = Size,
    y = TotalSales
  )
) +
  geom_point(
    color = "#dd7863",
    size = 3
  ) +
  geom_smooth(
    method = "lm",
    color = "#e9b447"
  ) +
  labs(
    title = "Store Size and Total Sales",
    x = "Store Size",
    y = "Total Sales ($)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



According to the scatter plot, there is a clear, strong positive linear relationship between store size and total sales. Larger stores consistently generate significantly higher cumulative sales.

Variances in total sales widen as store size grows. Stores with a size around 200,000 square feet show a wider spread in performance compared to smaller stores.

Store Type

Comparing store types can help determine whether certain categories of stores have consistently different average sales.

```
type_sales <- retail %>%
  group_by(Type) %>%
  summarize(
    AverageSales = mean(
      Weekly_Sales,
      na.rm = TRUE
    )
  )

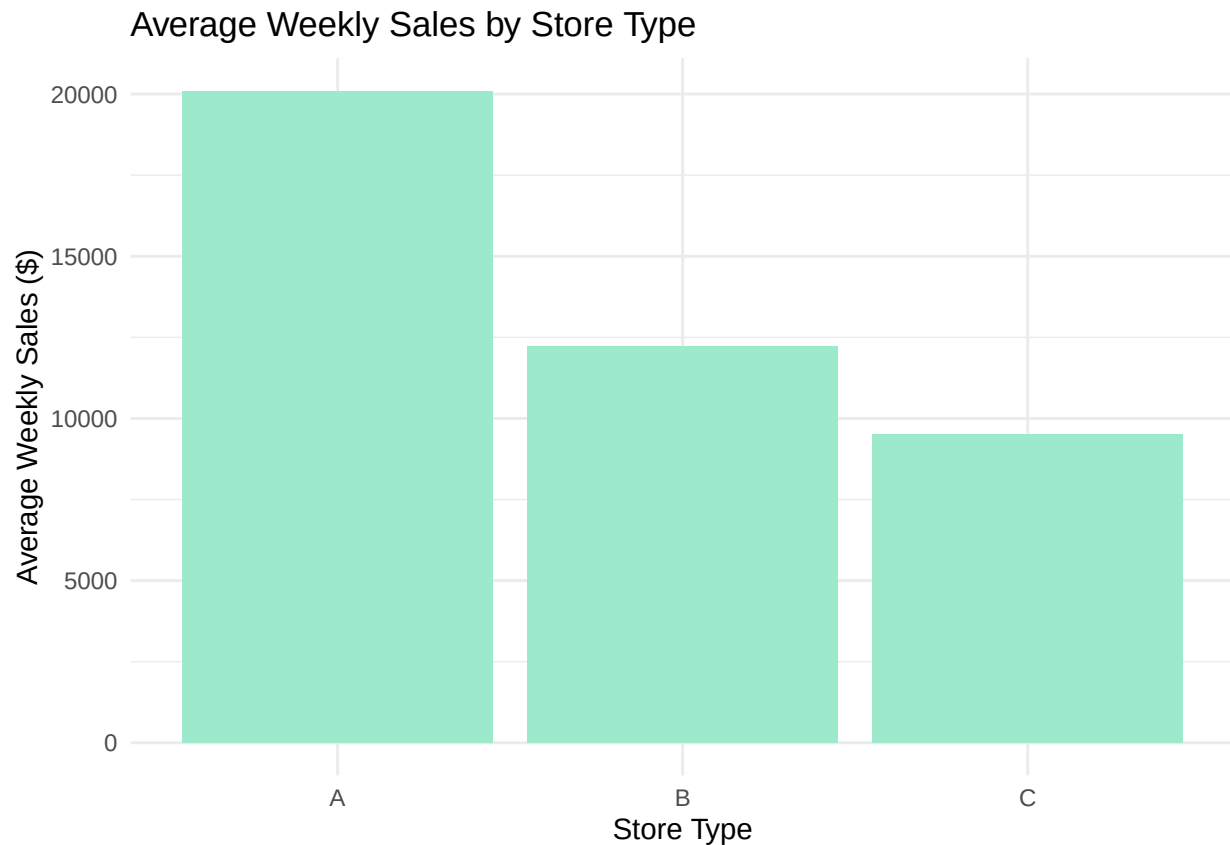
ggplot(
  type_sales,
  aes(
    x = Type,
    y = AverageSales
  )
) +
  geom_col(fill = "#9ce9cb") +
  labs(
```



```

title = "Average Weekly Sales by Store Type",
x = "Store Type",
y = "Average Weekly Sales ($)"
) +
theme_minimal()

```



Store Type A generates the highest average weekly sales at roughly \$20,000 followed by Type B at \$12,200 and Type C with the lowest average \$9,500.

Predictive Modeling

The exploratory analysis identifies several variables that may be associated with weekly sales. A linear regression model can be used to evaluate several predictors simultaneously.

```

model <- lm(
  Weekly_Sales ~
    IsHoliday +
    Temperature +
    Fuel_Price +
    CPI +
    Unemployment +
    TotalMarkdown,
  data = retail
)
summary(model)

```

```

##
## Call:

```

```
## lm(formula = Weekly_Sales ~ IsHoliday + Temperature + Fuel_Price +
##     CPI + Unemployment + TotalMarkdown, data = retail)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -32103 -14513  -8960   4857 600511
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.792e+04  1.370e+03  20.388 < 2e-16 ***
## IsHolidayTRUE -8.049e+02  2.761e+02  -2.915  0.00356 **
## Temperature   5.800e+01  4.781e+00  12.131 < 2e-16 ***
## Fuel_Price    -1.637e+03  3.346e+02  -4.892 9.98e-07 ***
## CPI           -2.640e+01  2.309e+00 -11.431 < 2e-16 ***
## Unemployment  -7.086e+02  4.941e+01 -14.341 < 2e-16 ***
## TotalMarkdown  1.048e-01  3.927e-03  26.696 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24600 on 97049 degrees of freedom
## (324514 observations deleted due to missingness)
## Multiple R-squared:  0.01095,    Adjusted R-squared:  0.01089
## F-statistic: 179.1 on 6 and 97049 DF,  p-value: < 2.2e-16
```

```
model_results <- data.frame(
  Variable = names(coef(model)),
  Coefficient = coef(model),
  P_Value = summary(model)$coefficients[, 4]
)
```

```
model_results
```

	Variable	Coefficient	P_Value
##	(Intercept)	27922.2999795	3.367783e-92
##	IsHolidayTRUE	-804.8567736	3.557790e-03
##	Temperature	57.9969959	7.632669e-34
##	Fuel_Price	-1637.0462025	9.976122e-07
##	CPI	-26.3967454	3.051743e-30
##	Unemployment	-708.5632933	1.357934e-46
##	TotalMarkdown	0.1048236	1.923144e-156

The model shows how the other variables are associated with sales while holding other variables constant.

The Temperature coefficient shows that with every 1 degree of Fahrenheit increase, sales increase by about \$58. With the Fuel Price, sales decrease by about \$1637 every dollar increase in Fuel Price. Also, when the Unemployment rate increases by every one percent, sales decrease about \$708. Holiday weeks are associated with about \$805 lower sales, holding the other variables constant. \$0.1048 increase per \$1 increase in total markdown.

The Holiday insight is interesting because the previous graph showed that Holidays tend to drive better sales. This differs from a simple comparison of holiday and non-holiday sales because the regression accounts for several other factors that may influence sales.

The model only has R^2 of 0.01095, so these variables explain only about 1.1% of the variation in weekly sales. This shows that the economic and promotional variables available in this data set do not explain much of the store-level variation in weekly sales.

Conclusion

This analysis examined the factors associated with weekly retail sales by combining sales data with information on economic conditions, promotional markdowns, holidays, and store characteristics. The exploratory analysis showed that sales vary across stores and that holiday periods, store size, and store type can be useful when understanding differences in retail performance.

The linear regression model found that all of the included variables were statistically significant, but the model explained only about 1.1% of the variation in weekly sales. This suggests that many other factors not included in the model, such as individual store and department characteristics, may play a much larger role in determining sales.

Working in retail, weekly sales are influenced more by employee behavior and store level operations than economic conditions. The way employees interact with the customer shapes their experiences and ultimately influences sales. Employee engagement, customer service, product knowledge, marketing displays, checkout efficiency, merchandising, and how well employees respond to customer needs can all affect how customers interact with a store. These factors are not captured in the current data set, but they can have a meaningful impact on daily and weekly performance.

These factors are difficult to capture in the available data set because it records the outcome of sales rather than behavior and the individual interactions that contribute to those sales. Two stores may experience the same economic conditions, have the same markdowns, and sell the same products, but still generate different sales because of differences in employee engagement, staffing, and how effectively employees communicate current promotions or selling priorities.

Overall, this project demonstrates how data science can be used to explore economic and business questions using real-world retail data. By combining exploratory data analysis, visualization, correlation analysis, and regression modeling, the analysis provides a better understanding of the factors associated with retail sales while also highlighting the limitations of using a small set of economic and promotional variables to predict complex consumer spending patterns.